

Deploy Unity on VR



how to deploy project on vr headset

Deploying a Unity project to a VR headset depends on **which headset you're using**, but the overall flow is similar. I'll give you a **clean step-by-step setup for Unity 2022.3+ (works with Unity 6.x too)** and cover the most common headsets like **Meta Quest (Quest 2/3/Pro)**.

How to Deploy Unity Project on VR Headset (Meta Quest Example)

1. Install Required Unity Modules

Open **Unity Hub** → **Installs** → (your Unity version)

Click **Add Modules** and install:

-  Android Build Support
-  Android SDK & NDK Tools
-  OpenJDK

 VR headsets like Quest run on **Android**, so this is mandatory.

2. Install XR Packages in Unity

Go to:

Window → **Package Manager**

Install:

- **XR Plugin Management**
- **OpenXR Plugin**
- (Optional but recommended) **XR Interaction Toolkit**

3. Enable XR in Project Settings

Go to:

Edit → Project Settings → XR Plug-in Management

Then:

- Click **Android tab**
- Enable:
✓ OpenXR

If asked, enable:

- ✓ Oculus / Meta Quest feature group (if available)
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✓ 4. Configure OpenXR

Go to:

Project Settings → XR Plug-in Management → OpenXR

Enable:

- ✓ Meta Quest Support (if visible)
 - ✓ Hand Tracking (optional)
 - ✓ Controller Profile
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✓ 5. Switch Build Platform to Android

Go to:

File → Build Settings

- Select **Android**
 - Click 🖱️ **Switch Platform**
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✓ 6. Player Settings (Important)

In Build Settings → **Player Settings**

Set:

Other Settings:

- Scripting Backend → **IL2CPP**

- Target Architectures → ✓ ARM64
 - Minimum API Level → Android 10 or higher
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✓ 7. Set XR Origin in Scene

In your scene:

GameObject → **XR** → **XR Origin (Action-based)**

Also add:

- XR Interaction Manager

👉 This gives VR camera + controllers.

✓ 8. Connect VR Headset (Meta Quest)

Option A: Developer Mode (USB cable)

1. Enable Developer Mode in Meta app (mobile)
2. Connect Quest to PC via USB
3. Put headset on → allow USB debugging

Option B: Wireless (Air Link)

- Enable Air Link in headset
 - Connect via same Wi-Fi network
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✓ 9. Build & Run

Go to:

File → **Build Settings**

Click:

- ✓ Add Open Scenes
- ✓ Build And Run

Unity will:

- Compile APK
- Install directly into headset

If Using Meta Quest Only (Recommended Setup)

Install these:

- Meta Quest Developer Hub (optional but helpful)
- SideQuest (for APK sideloading)

Common Problems & Fixes

"Device not found"

- ✓ Enable USB debugging in headset
- ✓ Use original USB cable
- ✓ Replug and accept permission popup in headset

Black screen in VR

- ✓ Check OpenXR enabled
- ✓ Ensure XR Origin is in scene
- ✓ Make sure camera is NOT Main Camera only

Controls not working

- ✓ Install XR Interaction Toolkit
- ✓ Use Action-based XR Origin

Quick Summary Flow

Unity Setup → Install Android + XR → Enable OpenXR → Switch Android → Add XR Origin → Connect Quest → Build & Run

If you want next level help:

I can guide you step-by-step for:

-  Hand tracking VR setup
-  Teleport movement system
-  VR interaction (grab objects, UI)
-  AR + VR hybrid project
-  Unity 6 VR template setup (clean + beginner friendly)

Just tell me 👍